

## Appendix 3: Domain Expert Review Guidance for the Blockchain-Based Design Pattern Decision Model

### Part I: Model Completeness

1. **Relevance of Design Patterns:** In your expert opinion, how relevant do you find the design patterns identified in the systematic literature review (SLR) for blockchain-based systems decision support? Are there elements that need to be added?

2. **Completeness of the Decision Model:** Is the content of the decision model sufficient? What is its level of coverage for blockchain system architecture?

3. **Uneven Distribution of the Decision Model:** What do you think about the uneven distribution of design patterns among different layers, such as the consensus layer, the network layer, and the incentive layer?

*Do you consider consensus layer algorithms design patterns? What do you think about practical solutions not being recognized as design patterns for network and incentive layers?*

### Part II: Model Correctness

4. **Suitability of Quality Attributes:** Could you provide your insights on whether the identified quality attribute metrics accurately and comprehensively describe all of the collected design patterns' goals and consequences?

5. **Rational of Concerns and Challenges:** In your experience, how do you consider the identified concerns and typical challenges commonly encountered in blockchain-based systems? Additionally, how effective do you find design patterns in providing corresponding solutions for these typical challenges?

6. **Accuracy of Mapping Associations:** What are your opinions on the accuracy and effectiveness of the relationships between quality attributes and design patterns?

### **Part III: Model Usefulness**

7. **Guidance for System Development:** For a blockchain application under design and development, can our model help in decision-making? If so, which aspects specifically can it help?

8. **Guidance for System Evaluation:** For an existing blockchain application, can our model help evaluate the application attributes? If so, what are the specific aspects that can be reflected?

9. **Understandability of the Decision Model:** Do you think this decision model is easy to understand and convenient for you to use? And Is it good for a novice to get started?

10. Do you have any other suggestions for our work? And are there any other opinions you would like to add?